21 - Fitting a Linear Model with OLS. Decomposition of Variance. Coefficient of Determination. Properties of the Parameter Estimators Without and With the Assumption of Gaussianity of the Error Term in the Model

Applied Statistics

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1 The Big Picture

- We observe n measurements of some outcome y (e.g. house prices) and r features (e.g. size, age) collected in a matrix $\mathbb{Z}$.
- **Goal:** Find the best coefficients $\beta_0, \beta_1, \ldots, \beta_r$ such that $y \approx \mathbb{Z}\beta$.
- We want to know:
  - How close can we get? $\to$ **OLS / Projection**
  - How good is the fit? $\to$ R^2
  - How reliable are the estimated coefficients? $\to$ **Properties of $\underline{\hat{\beta}}$**
  - Can we do hypothesis tests and build confidence intervals? $\to$ **Gaussian assumption**
- The road map: **Model** $\to$ **OLS = projection** $\to$ **Hat matrix H** $\to$ **Decomposition $y = \hat{y} + \hat{\epsilon}$** $\to$ R^2 $\to$ **Properties of $\hat{\beta}$, $\hat{\epsilon}$** $\to$ **Gaussian assumption** $\to$ **Full inference**.

2 The Model and Notation

- The **linear model** is:

$$\underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon} \tag{1}$$

- Dimensions:
  - $\underline{y} \in \mathbb{R}^n$: vector of n observations (the outcome we measure).

- $\mathbb{Z} \in \mathbb{R}^{n\times(r+1)}$: design matrix. Each row is one observation; each column is one feature. The first column is all ones (for the intercept β_0).
- $\underline{\beta} \in \mathbb{R}^{r+1}$: coefficient vector we want to estimate.
- $\underline{\epsilon} \in \mathbb{R}^n$: error term — random noise we cannot control.

- **Assumptions on the error:**
  - $\mathbb{E}[\underline{\epsilon}] = \underline{0}$ (zero mean: no systematic bias)
  - $\mathrm{Cov}(\underline{\epsilon}) = \sigma^2 I$ (errors are uncorrelated and have equal variance σ^2)
- **Note:** $\mathrm{Cov}(\underline{\epsilon}) = \sigma^2 I$ means the identity matrix is the right metric → Euclidean distance is the correct distance to minimize. If the covariance structure were different (correlated or unequal variance), we would use Mahalanobis distance → Generalized Least Squares.

3 Ordinary Least Squares (OLS)

3.1 The Optimization Problem

- **Goal:** Find $\hat{\underline{\beta}}$ that minimizes the total squared error between predictions and observations:

$$\hat{\underline{\beta}} = \arg\min_{\underline{\beta}\in\mathbb{R}^{r+1}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 = \arg\min_{\underline{\beta}} \sum_{i=1}^{n} (y_i - z_i^T \underline{\beta})^2 \tag{2}$$

- **Why squared errors?** Positive and negative errors do not cancel; large errors are penalized more heavily.
- **Solution** (when $\mathbb{Z}$ is full rank):

$$\boxed{\hat{\underline{\beta}} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y}} \tag{3}$$

- This is a *closed-form* analytical solution — no iterative optimization needed.
- $\hat{\underline{\beta}}$ is a **linear transformation** of $\underline{y}$: just a fixed matrix applied to the data.

3.2 Geometric Interpretation: OLS as Orthogonal Projection

- Think of $\mathbb{R}^n$ as a high-dimensional space. The vector $\underline{y}$ is a point in this space.
- The columns of $\mathbb{Z}$ span a subspace $\mathcal{L}(\mathbb{Z})$ of dimension $r+1$ (a "flat" embedded in $\mathbb{R}^n$). Any prediction $\mathbb{Z}\underline{\beta}$ *must* lie inside $\mathcal{L}(\mathbb{Z})$.
- **Key question:** Which point inside $\mathcal{L}(\mathbb{Z})$ is closest to $\underline{y}$ in Euclidean distance?
- **Answer:** The **orthogonal projection** of $\underline{y}$ onto $\mathcal{L}(\mathbb{Z})$:

$$\hat{\underline{y}} = \pi_{\underline{y}|\mathcal{L}(\mathbb{Z})} = \mathbb{Z}\hat{\underline{\beta}} \tag{4}$$

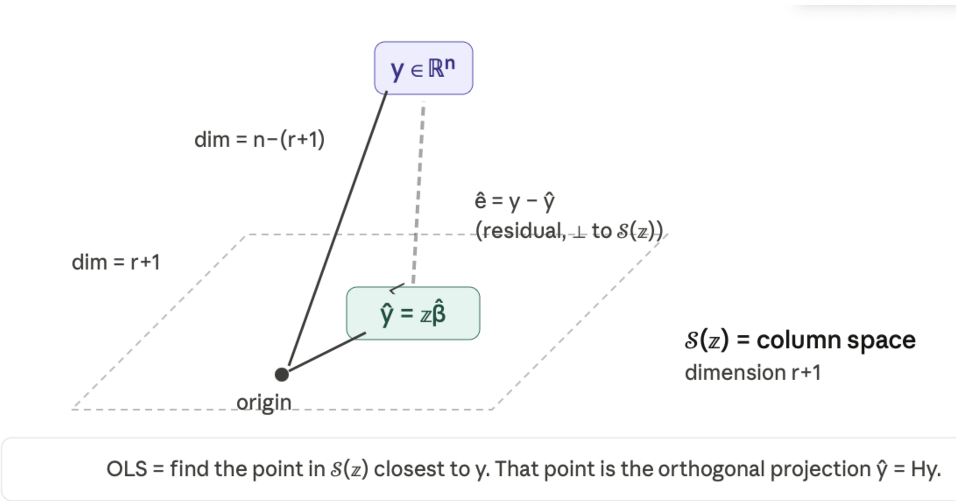

Figure 1: Projection

3.3 The Hat Matrix

- The **fitted values** are:

$$\hat{\underline{y}} = \mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y} = H\underline{y} \tag{5}$$

- The matrix $H := \mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T$ is called the **Hat Matrix**: it "puts a hat on $\underline{y}$", transforming observations into fitted values.
- H is the orthogonal projector onto $\mathcal{L}(\mathbb{Z})$: it is symmetric ($H^T = H$) and idempotent ($H^2 = H$).
- $I - H$ is the orthogonal projector onto $\mathcal{L}^{\perp}(\mathbb{Z})$ (also symmetric and idempotent).

3.4 Proof Sketch: Why $\hat{\beta} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y}$

- Since $\mathbb{Z}$ is full rank, $\mathbb{Z}^T\mathbb{Z}$ is an $(r+1)\times(r+1)$ invertible matrix. Take its spectral decomposition:

$$\mathbb{Z}^T\mathbb{Z} = \sum_{i=1}^{r+1} \lambda_i \underline{e}_i \underline{e}_i^T \quad \text{with } \lambda_1 \geq \cdots \geq \lambda_{r+1} > 0 \tag{6}$$

- Its inverse is:

$$(\mathbb{Z}^T\mathbb{Z})^{-1} = \sum_{i=1}^{r+1} \lambda_i^{-1} \underline{e}_i \underline{e}_i^T \tag{7}$$

- Define $\underline{q}_i = \frac{1}{\sqrt{\lambda_i}}\mathbb{Z}\underline{e}_i$ for $i = 1, \ldots, r+1$. Then:
  - Each $\underline{q}_i \in \mathcal{L}(\mathbb{Z})$ by definition.
  - They are orthonormal: $\underline{q}_i^T \underline{q}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$

- So $\{\underline{q}_1, \ldots, \underline{q}_{r+1}\}$ is an orthonormal basis for $\mathcal{L}(\mathbb{Z})$. Projecting $\underline{y}$:

$$\pi_{\underline{y}|\mathcal{L}(\mathbb{Z})} = \sum_{i=1}^{r+1} \underline{q}_i \frac{\underline{q}_i^T \underline{y}}{\underline{q}_i^T \underline{q}_i} = \mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T \underline{y} = \hat{\underline{y}} \tag{8}$$

- **Note (non-full rank case):** If $\text{rank}(\mathbb{Z}) = k < r + 1$, use the Moore-Penrose pseudo-inverse:

$$(\mathbb{Z}^T\mathbb{Z})^\dagger = \sum_{i=1}^{k} \lambda_i^{-1} \underline{e}_i \underline{e}_i^T \tag{9}$$

The projection $\hat{\underline{y}}$ is still unique, but $\hat{\underline{\beta}}$ is not (infinitely many solutions).

4 Fitted Values, Residuals, and the Pythagorean Decomposition

4.1 Definitions

- **Fitted values:**

$$\hat{\underline{y}} = H\underline{y} = \pi_{\underline{y}|\mathcal{L}(\mathbb{Z})} \tag{10}$$

What the model can explain from the features.

- **Residuals:**

$$\hat{\underline{\epsilon}} = \underline{y} - \hat{\underline{y}} = (I - H)\underline{y} = \pi_{\underline{y}|\mathcal{L}^\perp(\mathbb{Z})} \tag{11}$$

The part of $\underline{y}$ the model cannot capture.

- Always: $\hat{\underline{\epsilon}} \perp \hat{\underline{y}}$ (residuals are perpendicular to fitted values by construction of projection).

4.2 Pythagorean Decomposition

- Since $\underline{y} = \hat{\underline{y}} + \hat{\underline{\epsilon}}$ and $\hat{\underline{y}} \perp \hat{\underline{\epsilon}}$, Pythagoras gives:

$$\|\underline{y}\|^2 = \|\hat{\underline{y}}\|^2 + \|\hat{\underline{\epsilon}}\|^2 \implies \sum_{i=1}^{n} y_i^2 = \sum_{i=1}^{n} \hat{y}_i^2 + \sum_{i=1}^{n} \hat{\epsilon}_i^2 \tag{12}$$

- In shorthand: $SS_{tot} = SS_{reg} + SS_{res}$.
- Total variation = explained + unexplained.

4.3 Dimensions and Overfitting

- $\dim(\mathcal{L}(\mathbb{Z})) = r + 1$ (model space).
- $\dim(\mathcal{L}^\perp(\mathbb{Z})) = n - (r + 1)$ (residual space).
- If $r = n - 1$: the model fills all of $\mathbb{R}^n \to \hat{\underline{\epsilon}} = \underline{0} \to$ perfect fit, but **zero prediction power**: **overfitting**.

4.4 Important Remark on Residuals

- $\hat{\underline{\epsilon}}$ is **not** an estimate of the true error $\underline{\epsilon}$.
- The true errors $\underline{\epsilon}$ are random and live in all of $\mathbb{R}^n$.
- The residuals $\hat{\underline{\epsilon}}$ are *constrained* to $\mathcal{L}^{\perp}(\mathbb{Z})$ — a proper subspace. It is a deterministic function of the data; it is a proxy, not a realisation.

5 Coefficient of Determination (R^2)

5.1 Variance Decomposition Around the Mean

- Assume $\underline{1} \in \mathcal{L}(\mathbb{Z})$ (the design matrix includes an intercept — its first column is all ones).
- Projecting $\underline{y}$ onto $\mathcal{L}(\{\underline{1}\})$ gives $\bar{y} \cdot \underline{1}$ (the mean vector).
- The **centred sum of squares decomposition** is:

$$\underbrace{\sum_{i=1}^{n}(y_i - \bar{y})^2}_{CSS_{tot}} = \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{CSS_{reg}} + \underbrace{\sum_{i=1}^{n}\hat{\epsilon}_i^2}_{SS_{res}} \tag{13}$$

  equivalently: $CSS_{tot} = CSS_{reg} + SS_{res}$.
- Total variability = variability explained by the model + residual variability.

5.2 Definition of R^2

- The **coefficient of determination** is:

$$R^2 = 1 - \frac{SS_{res}}{CSS_{tot}} = 1 - \frac{\sum_{i=1}^{n}\hat{\epsilon}_i^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \sin^2(\theta) = \frac{CSS_{reg}}{CSS_{tot}} \tag{14}$$

  where θ is the angle between $\hat{\underline{\epsilon}}$ and $\hat{\underline{y}} - \bar{y} \cdot \underline{1}$.
- R^2 is the proportion of total variability explained by the regression model.
- Interpretation:
  - $R^2 = 1$: perfect fit ($\theta = 0$, $\hat{\underline{\epsilon}} = \underline{0}$). Risk of overfitting if $r + 1 = n$.
  - $R^2 = 0$: model only predicts the mean ($\theta = \pi/2$, $\hat{y}_i = \bar{y}\ \forall i$).
  - $R^2 \in [0, 1]$ only when $\underline{1} \in \mathcal{L}(\mathbb{Z})$; otherwise it can be negative.

5.3 Regression Through the Origin

- If $\beta_0 = 0$ is forced, then $\underline{1} \notin \mathcal{L}(\mathbb{Z})$ and R^2 is no longer a valid proportion.
- Use instead:

$$\tilde{R}^2 = 1 - \frac{||\hat{\underline{\epsilon}}||^2}{||\underline{y}||^2} \tag{15}$$

  This is a measure of fit, not a proportion of variance.
- **Note:** We can make $\underline{1} \in \mathcal{L}(\mathbb{Z})$ even with regression through the origin by appropriate construction of $\mathbb{Z}$.

5.4 Adjusted R^2

- R^2 always increases when adding more features (even useless ones), since a bigger space projects closer to $\underline{y}$.
- To penalize model complexity, normalize each sum by its **degrees of freedom**:

$$R^2_{adj} = 1 - \frac{\frac{1}{n-(r+1)} \sum_{i=1}^{n} \hat{\epsilon}_i^2}{\frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{16}$$

- Degrees of freedom:
  - Residuals: $n-(r+1)$ free components ($r+1$ are constrained by the model).
  - Total centred variability: $n-1$ free components (constrained by the mean).
- If $r+1=n$: R^2_{adj} is undefined $(0/0)$ — the overfitting regime.
- R^2_{adj} encodes the **bias-variance trade-off**: penalizing additional parameters that do not reduce residuals enough.

6 Properties of $\underline{\hat{\beta}}$ and $\underline{\hat{\epsilon}}$ (No Gaussian Assumption)

Assume $\text{rank}(\mathbb{Z}) = r+1 \leq n$ and $\text{Cov}(\underline{\epsilon}) = \sigma^2 I$.

6.1 Theorem

1. $\mathbb{E}[\underline{\hat{\beta}}] = \underline{\beta} \quad$ ($\underline{\hat{\beta}}$ is **unbiased**)
2. $\text{Cov}(\underline{\hat{\beta}}) = \sigma^2 (\mathbb{Z}^T \mathbb{Z})^{-1}$
3. $\mathbb{E}[\underline{\hat{\epsilon}}] = \underline{0}$
4. $\text{Cov}(\underline{\hat{\epsilon}}) = \sigma^2 (I - H)$
5. $\mathbb{E}[\underline{\hat{\epsilon}}^T \underline{\hat{\epsilon}}] = \sigma^2 (n-(r+1))$

6.2 Proofs

- **Proof of 1:**
$$\mathbb{E}[\underline{\hat{\beta}}] = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\mathbb{E}[\underline{y}] = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\mathbb{Z}\underline{\beta} = \underline{\beta}$$
- **Proof of 2:**
$$\text{Cov}(\underline{\hat{\beta}}) = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T \cdot \sigma^2 I \cdot \mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1} = \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}$$
- **Proof of 3:**
$$\mathbb{E}[\underline{\hat{\epsilon}}] = (I-H)\mathbb{E}[\underline{y}] = (I-H)\mathbb{Z}\underline{\beta} = \mathbb{Z}\underline{\beta} - \mathbb{Z}\underline{\beta} = \underline{0}$$
since $H\mathbb{Z} = \mathbb{Z}$ (columns of $\mathbb{Z}$ are already in $\mathcal{L}(\mathbb{Z})$).

- **Proof of 4:** Since $I - H$ is a symmetric idempotent projector:

$$\mathrm{Cov}(\hat{\underline{\epsilon}}) = (I-H)\sigma^2 I(I-H)^T = \sigma^2(I-H)^2 = \sigma^2(I-H)$$

- **Proof of 5 (trace trick):** A scalar equals its own trace, so:

$$\mathbb{E}[\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}] = \mathrm{tr}\big(\mathbb{E}[\hat{\underline{\epsilon}}\hat{\underline{\epsilon}}^T]\big) = \mathrm{tr}(\sigma^2(I-H)) = \sigma^2(n - \mathrm{tr}(H))$$

Now $\mathrm{tr}(H) = \mathrm{tr}(\mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T) = \mathrm{tr}(\mathbb{Z}^T\mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}) = \mathrm{tr}(I_{r+1}) = r+1$.

$$\therefore \quad \mathbb{E}[\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}] = \sigma^2(n-(r+1))$$

6.3 Unbiased Estimator of σ^2

- **Definition:**

$$S^2 = \frac{\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}}{n-(r+1)} = \frac{\|\hat{\underline{\epsilon}}\|^2}{n-(r+1)} \tag{17}$$

- **Corollary:** $\mathbb{E}[S^2] = \sigma^2$ (unbiased).
- The denominator $n-(r+1)$ is the **degrees of freedom** of the residuals: the number of free dimensions in the residual vector.
- **Design of Experiments:** Since $\mathrm{Cov}(\hat{\underline{\beta}}) = \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}$, we can choose $\mathbb{Z}$ to make $(\mathbb{Z}^T\mathbb{Z})^{-1}$ small, reducing uncertainty about $\hat{\underline{\beta}}$.

7 Properties Under the Gaussian Assumption

7.1 The Gaussian Assumption

- Add: $\underline{\epsilon} \sim \mathcal{N}_n(\underline{0}, \sigma^2 I)$.
- This implies: $\underline{y} \sim \mathcal{N}_n(\mathbb{Z}\underline{\beta}, \sigma^2 I)$.
- This assumption allows us to derive exact distributions and perform full statistical inference.

7.2 Theorem: Properties Under Gaussianity

Assume $\underline{\epsilon} \sim \mathcal{N}_n(\underline{0}, \sigma^2 I)$ and $\mathrm{rank}(\mathbb{Z}) = r+1 \leq n$. Then:

1. $\hat{\underline{\beta}}$ and $\hat{\sigma}^2 = \dfrac{\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}}{n}$ are the **maximum likelihood estimators** of $\underline{\beta}$ and σ^2.
2. $\hat{\underline{\beta}} \sim \mathcal{N}_{r+1}\big(\underline{\beta},\ \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}\big)$
3. $\hat{\underline{\epsilon}} \sim \mathcal{N}_n(\underline{0},\ \sigma^2(I-H))$ (**singular Gaussian**, supported on $\mathcal{L}^{\perp}(\mathbb{Z})$)
4. $\hat{\underline{\epsilon}} \perp\!\!\!\perp \hat{\underline{\beta}}$ (**independent**)
5. $\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}} \sim \sigma^2\chi^2(n-(r+1))$

7.3 Why is $\underline{\hat{\epsilon}}$ a Singular Gaussian?

- The matrix $I - H$ has rank $n - (r+1) < n$, so $\det(I - H) = 0$.
- This is a Gaussian distribution whose support is a proper subspace of $\mathbb{R}^n$ (Lebesgue measure zero in $\mathbb{R}^n$).
- The true errors $\underline{\epsilon}$ are non-singular Gaussians; the residuals $\underline{\hat{\epsilon}}$ are constrained to live in $\mathcal{L}^{\perp}(\mathbb{Z})$.
- To compute $\underline{\hat{\epsilon}}^T(I-H)^{-1}\underline{\hat{\epsilon}}$ we use the Moore-Penrose inverse: $\underline{\hat{\epsilon}}^T(I-H)^{\dagger}\underline{\hat{\epsilon}} \sim \sigma^2\chi^2(n-(r+1))$.

7.4 Proof of Independence (Property 4)

- Write the joint vector as a linear map of $\underline{y}$:

$$\begin{pmatrix} \underline{\hat{\beta}} \\ \underline{\hat{\epsilon}} \end{pmatrix} = \underbrace{\begin{pmatrix} (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T \\ I - \mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T \end{pmatrix}}_{A} \underline{y} \tag{18}$$

- Compute the joint covariance $\sigma^2 AA^T$:

$$AA^T = \begin{pmatrix} (\mathbb{Z}^T\mathbb{Z})^{-1} & 0 \\ 0 & I - H \end{pmatrix} \tag{19}$$

- The off-diagonal blocks are zero $\rightarrow$ in a Gaussian, zero covariance = independence.

7.5 The Pivotal Quantity

- Since $\underline{\hat{\beta}} \sim \mathcal{N}_{r+1}(\underline{\beta}, \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1})$, its Mahalanobis distance from the true $\underline{\beta}$:

$$\frac{1}{\sigma^2}(\underline{\hat{\beta}} - \underline{\beta})^T(\mathbb{Z}^T\mathbb{Z})(\underline{\hat{\beta}} - \underline{\beta}) \sim \chi^2(r+1) \tag{20}$$

- Independently, $\underline{\hat{\epsilon}}^T\underline{\hat{\epsilon}} \sim \sigma^2\chi^2(n-(r+1))$.
- Their ratio (dividing each by its degrees of freedom) gives an F-distribution, and crucially σ^2 cancels:

$$\boxed{(\underline{\hat{\beta}} - \underline{\beta})^T(\mathbb{Z}^T\mathbb{Z})(\underline{\hat{\beta}} - \underline{\beta})\,\frac{1}{S^2(r+1)} \sim F(r+1,\ n-(r+1))} \tag{21}$$

- This is the **pivotal quantity**: it has a known distribution and contains no unknown parameters. All inference is built from this.

8 Inference: Confidence Regions and Intervals

8.1 Confidence Region for $\underline{\beta}$

- The $1-\alpha$ confidence region for $\underline{\beta}$ is an ellipse centred at $\hat{\underline{\beta}}$:

$$CR_{1-\alpha}(\underline{\beta}) = \left\{ \underline{\beta} \in \mathbb{R}^{r+1} : \frac{1}{S^2}(\hat{\underline{\beta}} - \underline{\beta})^T (\mathbb{Z}^T\mathbb{Z})(\hat{\underline{\beta}} - \underline{\beta}) \leq (r+1)F_\alpha(r+1,\, n-(r+1)) \right\} \tag{22}$$

- $(1-\alpha) \times 100\%$ of the time, the true $\underline{\beta}$ lies inside this ellipse.

8.2 Confidence Interval for σ^2

- Since $(n-(r+1))S^2/\sigma^2 \sim \chi^2(n-(r+1))$:

$$CI_{1-\alpha}(\sigma^2) = \left[(n-(r+1))S^2 \frac{1}{\chi^2_{\alpha/2}(n-(r+1))}, \quad (n-(r+1))S^2 \frac{1}{\chi^2_{1-\alpha/2}(n-(r+1))} \right] \tag{23}$$

8.3 Simultaneous Confidence Interval for $\underline{a}^T\underline{\beta}$

- For any linear combination $\underline{a}^T\underline{\beta}$ with $\underline{a} \in \mathbb{R}^{r+1}$, using the **Maximum Lemma**:

$$\max_{\underline{a}\in\mathbb{R}^{r+1}} \frac{1}{S^2} \frac{(\underline{a}^T(\hat{\underline{\beta}} - \underline{\beta}))^2}{\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}} = \frac{1}{S^2}(\hat{\underline{\beta}} - \underline{\beta})^T(\mathbb{Z}^T\mathbb{Z})(\hat{\underline{\beta}} - \underline{\beta}) \sim (r+1)F(r+1, n-(r+1)) \tag{24}$$

- The simultaneous confidence interval is:

$$SimCI_{1-\alpha}(\underline{a}^T\underline{\beta}) = \left\{ \underline{a}^T\hat{\underline{\beta}} \pm \sqrt{\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}} \cdot \sqrt{S^2(r+1)F_{1-\alpha}(r+1,\, n-(r+1))} \right\} \tag{25}$$

Türkçe Özet (Turkish Summary)

1. Büyük Resim: Ne Yapmaya Çalışıyoruz?

- n gözlemimiz var (örn. 100 evin fiyatı $\to \underline{y}$) ve r özelliğimiz var (örn. m^2, oda sayısı $\to \mathbb{Z}$).
- **Hedef:** En iyi $\beta_0, \beta_1, \ldots, \beta_r$ katsayılarını bul:

$$\text{Fiyat} \approx \beta_0 + \beta_1 \cdot \text{m}^2 + \beta_2 \cdot \text{oda} + \cdots$$

- Cevaplanacak sorular:
  - En iyi fit nasıl bulunur? $\to$ **OLS / Projeksiyon**
  - Fit ne kadar iyi? $\to R^2$
  - Katsayılar ne kadar güvenilir? $\to$ **$\hat{\underline{\beta}}$'nın özellikleri**
  - Test ve güven aralığı yapabilir miyiz? $\to$ **Gaussian varsayımı**
- Yol haritası: **Model** $\to$ **OLS = projeksiyon** $\to$ **Hat matrisi** $H \to y = \hat{y} + \hat{\epsilon}$ **ayrışımı** $\to R^2 \to \hat{\beta}$**,** $\hat{\epsilon}$ **özellikleri** $\to$ **Gaussian** $\to$ **Tam çıkarım**.

2. Model ve Notasyon

- **Lineer model:**

$$\underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon}$$

- Boyutlar:
  - $\underline{y} \in \mathbb{R}^n$: n gözlemden oluşan vektör (tahmin etmek istediğimiz şey).
  - $\mathbb{Z} \in \mathbb{R}^{n\times(r+1)}$: tasarım matrisi. Her satır bir gözlem, her sütun bir özellik. İlk sütun tamamen 1'lerden oluşur (β_0 sabiti için).
  - $\underline{\beta} \in \mathbb{R}^{r+1}$: tahmin etmek istediğimiz katsayı vektörü.
  - $\underline{\epsilon} \in \mathbb{R}^n$: hata terimi — kontrol edemediğimiz rastgele gürültü.
- **Hata varsayımları:**
  - $\mathbb{E}[\underline{\epsilon}] = \underline{0}$ (sıfır ortalama: sistematik yanlılık yok)
  - $\text{Cov}(\underline{\epsilon}) = \sigma^2 I$ (hatalar birbirinden bağımsız, hepsinin varyansı σ^2)
- **Not:** $\text{Cov}(\underline{\epsilon}) = \sigma^2 I$ olduğu için Öklid mesafesi doğru metriktir. Kovaryans yapısı farklı olsaydı (korelasyonlu ya da eşit olmayan varyans), Mahalanobis mesafesi kullanılırdı $\to$ Genelleştirilmiş En Küçük Kareler.

3. OLS: En İyi β'yı Nasıl Buluyoruz?

- **Amaç:** Tahminlerimiz ($\mathbb{Z}\underline{\beta}$) ile gerçek değerler ($\underline{y}$) arasındaki toplam karesel hatayı minimize et:

$$\hat{\underline{\beta}} = \arg\min_{\underline{\beta}\in\mathbb{R}^{r+1}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 = \arg\min_{\underline{\beta}} \sum_{i=1}^{n} (y_i - z_i^T \underline{\beta})^2$$

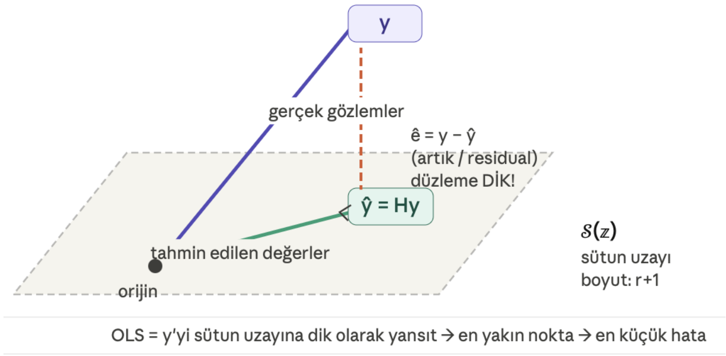

Figure 2: Projection

- **Neden karesini alıyoruz?** Pozitif ve negatif hatalar birbirini götürmesin; büyük hatalar daha çok cezalandırılsın.
- **Çözüm** ($\mathbb{Z}$ tam rank ise):
$$\underline{\hat{\beta}} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y}$$
- Bu kapalı form bir çözümdür — iterasyon gerektirmez. $\underline{\hat{\beta}}$, $\underline{y}$'nin doğrusal bir dönüşümüdür.

Geometrik Sezgi: OLS = Dik Projeksiyon

- $\mathbb{R}^n$'i yüksek boyutlu bir uzay olarak düşün. $\underline{y}$ bu uzayda bir noktadır.
- $\mathbb{Z}$'nin sütunları $r+1$ boyutlu bir alt uzay $\mathcal{L}(\mathbb{Z})$ oluşturur. Her tahmin $\mathbb{Z}\underline{\beta}$ bu alt uzayın içinde olmak zorundadır.
- **Soru:** $\mathcal{L}(\mathbb{Z})$ içindeki hangi nokta $\underline{y}$'ye en yakın?
- **Cevap:** Dik projeksiyon! $\underline{y}$'yi bu düzleme dik olarak "düşürürsün" — bu nokta $\underline{\hat{y}}$'dir.
- **Hat Matrisi:** $H = \mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T$ matrisi $\underline{y}$'ye "şapka giydiriyor":
$$\underline{\hat{y}} = H\underline{y}$$
H simetrik ve idempotanttır ($H^T = H$, $H^2 = H$). $I - H$ ise $\mathcal{L}^\perp(\mathbb{Z})$'ye dik projektördür.
- **Not (tam rank değilse):** Moore-Penrose sözde tersi kullan: $(\mathbb{Z}^T\mathbb{Z})^\dagger = \sum_{i=1}^k \lambda_i^{-1}\underline{e}_i\underline{e}_i^T$. Projeksiyon $\underline{\hat{y}}$ hâlâ tekil, ama $\underline{\hat{\beta}}$ artık tekil değil (sonsuz çözüm var).

4. Tahminler ve Artıklar: Pisagor Ayrışımı

- **Tahmin edilen değerler (fitted values):**
$$\underline{\hat{y}} = H\underline{y} = \pi_{\underline{y}|\mathcal{L}(\mathbb{Z})}$$
Modelin özellikleri (x) kullanarak $\underline{y}$ hakkında söyleyebildiği kısım.

- **Artıklar (residuals):**

$$\hat{\underline{\epsilon}} = \underline{y} - \underline{\hat{y}} = (I - H)\underline{y} = \pi_{\underline{y}|\mathcal{L}^{\perp}(\mathbb{Z})}$$

Modelin özelliklerden yakalayamadığı kısım.

- Her zaman: $\underline{\hat{y}} \perp \hat{\underline{\epsilon}}$ (projeksiyon tanımından gelir).
- **Pisagor Teoremi:** $\underline{y} = \underline{\hat{y}} + \hat{\underline{\epsilon}}$ ve ikisi dik $\Rightarrow$

$$\|\underline{y}\|^2 = \|\underline{\hat{y}}\|^2 + \|\hat{\underline{\epsilon}}\|^2 \implies \sum_{i=1}^{n} y_i^2 = \sum_{i=1}^{n} \hat{y}_i^2 + \sum_{i=1}^{n} \hat{\epsilon}_i^2$$

Kısaca: $SS_{tot} = SS_{reg} + SS_{res}$. Toplam varyasyon = açıklanan + açıklanamayan.

Boyutlar ve Overfitting

- $\dim(\mathcal{L}(\mathbb{Z})) = r + 1$ (model uzayı).
- $\dim(\mathcal{L}^{\perp}(\mathbb{Z})) = n - (r + 1)$ (artık uzayı).
- Eğer $r = n - 1$ ise: model tüm $\mathbb{R}^n$'i doldurur $\to \hat{\underline{\epsilon}} = \underline{0} \to$ mükemmel fit, ama **sıfır tahmin gücü: overfitting!**
- **Önemli not:** $\hat{\underline{\epsilon}}$, gerçek hata $\underline{\epsilon}$'un tahmini *değildir*. Gerçek hatalar $\mathbb{R}^n$'de rastgele yaşar; artıklar $\mathcal{L}^{\perp}(\mathbb{Z})$ alt uzayına hapsolmuştur. Bir gerçekleşme değil, bir vekil (proxy) dir.

5. R^2: Model Ne Kadar İyi?

- **Problem:** Fit iyiyse $\hat{\underline{\epsilon}}$ küçük olur — ama "küçük" ne demek? Neyle karşılaştırıyoruz?
- **Varyans ayrışımı** ($\underline{1} \in \mathcal{L}(\mathbb{Z})$ gerekli, yani tasarım matrisinde sabit terim var):

$$\underbrace{\sum_{i=1}^{n}(y_i - \bar{y})^2}_{CSS_{tot}} = \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{CSS_{reg}} + \underbrace{\sum_{i=1}^{n}\hat{\epsilon}_i^2}_{SS_{res}}$$

Toplam değişkenlik = modelin açıkladığı + modelin açıklayamadığı.

- **R^2 tanımı:**

$$R^2 = 1 - \frac{SS_{res}}{CSS_{tot}} = 1 - \sin^2(\theta) = \frac{CSS_{reg}}{CSS_{tot}}$$

θ, artık vektörü $\hat{\underline{\epsilon}}$ ile ortalamadan sapma vektörü $\underline{\hat{y}} - \bar{y} \cdot \underline{1}$ arasındaki açıdır.

- R^2, toplam varyasyonun modelle açıklanan oranıdır.
- **Yorumlar:**
  - $R^2 = 1$: mükemmel fit ($\theta = 0$, $\hat{\underline{\epsilon}} = \underline{0}$). Tehlike: $r + 1 = n$ ise overfitting!
  - $R^2 = 0$: model sadece ortalamayı tahmin ediyor ($\theta = \pi/2$, $\hat{y}_i = \bar{y}\ \forall i$). Kullanışsız.
  - $R^2 \in [0, 1]$ yalnızca $\underline{1} \in \mathcal{L}(\mathbb{Z})$ olduğunda. Aksi hâlde negatif olabilir.

Orijinden Regresyon

- $\beta_0 = 0$ zorlanırsa $\underline{1} \notin \mathcal{L}(\mathbb{Z}) \to R^2$ geçerli bir oran değil.
- Bunun yerine:
$$\tilde{R}^2 = 1 - \frac{\|\hat{\underline{\epsilon}}\|^2}{\|\underline{y}\|^2}$$
Bu bir fit ölçüsüdür, varyans oranı değil.
- **Not:** Orijinden regresyonda bile $\mathbb{Z}$'yi uygun şekilde kurarak $\underline{1} \in \mathcal{L}(\mathbb{Z})$ sağlanabilir.

Düzeltilmiş R^2

- R^2, işe yaramaz değişken eklenince bile her zaman artar (daha büyük uzay $\underline{y}$'ye daha yakın projeksiyon verir).
- Çözüm: Her toplamı kendi **serbestlik derecesine** böl:
$$R^2_{adj} = 1 - \frac{\dfrac{1}{n-(r+1)}\displaystyle\sum_{i=1}^{n}\hat{\epsilon}_i^2}{\dfrac{1}{n-1}\displaystyle\sum_{i=1}^{n}(y_i - \bar{y})^2}$$
- Serbestlik dereceleri:
  - Artıklar: $n-(r+1)$ serbest bileşen ($r+1$ tanesi model tarafından kısıtlanmış).
  - Toplam sapma: $n-1$ serbest bileşen (ortalama 1 kısıtlama koyuyor).
- Eğer $r+1 = n$ ise R^2_{adj} tanımsız $(0/0)$ — overfitting bölgesi.
- R^2_{adj}, **bias-variance trade-off**'u encode eder: artıkları yeterince azaltmayan ek parametreleri cezalandırır.

6. $\hat{\underline{\beta}}$ ve $\hat{\underline{\epsilon}}$'in Özellikleri (Gaussian Varsayımı Olmadan)

$\text{rank}(\mathbb{Z}) = r+1 \le n$ ve $\text{Cov}(\underline{\epsilon}) = \sigma^2 I$ varsayımlarıyla:

Teorem

1. $\mathbb{E}[\hat{\underline{\beta}}] = \underline{\beta} \to$ **tarafsız tahminleyici**
2. $\text{Cov}(\hat{\underline{\beta}}) = \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}$
3. $\mathbb{E}[\hat{\underline{\epsilon}}] = \underline{0}$
4. $\text{Cov}(\hat{\underline{\epsilon}}) = \sigma^2(I - H)$
5. $\mathbb{E}[\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}] = \sigma^2(n-(r+1))$

σ^2'nin Tarafsız Tahmincisi

- **Tanım:**
$$S^2 = \frac{\hat{\underline{\epsilon}}^T \hat{\underline{\epsilon}}}{n-(r+1)} = \frac{\|\hat{\underline{\epsilon}}\|^2}{n-(r+1)}, \qquad \mathbb{E}[S^2] = \sigma^2$$
- Payda $n-(r+1)$: artık vektörünün **serbestlik derecesi** — artık vektöründe kaç bağımsız bileşen olduğu.
- **Deney Tasarımı:** $\text{Cov}(\hat{\underline{\beta}}) = \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}$ olduğundan, $\mathbb{Z}$'yi öyle seçebiliriz ki $(\mathbb{Z}^T\mathbb{Z})^{-1}$ küçük olsun $\to$ $\hat{\underline{\beta}}$ daha kesin. Bu, Deney Tasarımı alanının temelidir.

7. Gaussian Varsayımı ve Tam İstatistiksel Çıkarım

Gaussian Varsayımı

- Ek varsayım: $\underline{\epsilon} \sim \mathcal{N}_n(\underline{0}, \sigma^2 I)$.
- Bu şunu zorunlu kılar: $\underline{y} \sim \mathcal{N}_n(\mathbb{Z}\underline{\beta}, \sigma^2 I)$.
- Bu varsayım sayesinde tam dağılımlar elde edebilir ve istatistiksel çıkarım yapabiliriz.

Teorem: Gaussian Altında Özellikler

$\underline{\epsilon} \sim \mathcal{N}_n(\underline{0}, \sigma^2 I)$ ve $\text{rank}(\mathbb{Z}) = r+1 \le n$ varsayımlarıyla:

1. $\hat{\underline{\beta}}$ ve $\hat{\sigma}^2 = \dfrac{\hat{\underline{\epsilon}}^T \hat{\underline{\epsilon}}}{n}$ **maksimum olabilirlik tahminleyicileri**dir.
2. $\hat{\underline{\beta}} \sim \mathcal{N}_{r+1}\big(\underline{\beta},\ \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}\big)$
3. $\hat{\underline{\epsilon}} \sim \mathcal{N}_n(\underline{0},\ \sigma^2(I-H))$ (**tekil Gaussian**, $\mathcal{L}^\perp(\mathbb{Z})$ üzerinde desteklenir)
4. $\hat{\underline{\epsilon}} \perp\!\!\!\perp \hat{\underline{\beta}}$ (**bağımsız!**)
5. $\hat{\underline{\epsilon}}^T \hat{\underline{\epsilon}} \sim \sigma^2 \chi^2(n-(r+1))$

Neden $\hat{\underline{\epsilon}}$ Tekil Gaussian?

- $I-H$ matrisinin rankı $n-(r+1) < n$, dolayısıyla $\det(I-H) = 0$.
- Bu, desteği $\mathbb{R}^n$'in bir alt uzayı olan (Lebesgue ölçüsü sıfır) bir Gaussian dağılımıdır.
- Gerçek hatalar $\underline{\epsilon}$: tekil olmayan Gaussian, tüm $\mathbb{R}^n$'de yaşar.
- Artıklar $\hat{\underline{\epsilon}}$: $\mathcal{L}^\perp(\mathbb{Z})$ alt uzayına kısıtlanmış tekil Gaussian.
- $\hat{\underline{\epsilon}}^T(I-H)^{-1}\hat{\underline{\epsilon}}$ hesaplamak için Moore-Penrose sözde tersi gerekir: $\hat{\underline{\epsilon}}^T(I-H)^{\dagger}\hat{\underline{\epsilon}} \sim \sigma^2\chi^2(n-(r+1))$.

Pivotal Miktar

- $\underline{\hat{\beta}} \sim \mathcal{N}_{r+1}(\underline{\beta}, \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1})$ olduğundan, gerçek $\underline{\beta}$'dan Mahalanobis uzaklığı:

$$\frac{1}{\sigma^2}(\underline{\hat{\beta}} - \underline{\beta})^T(\mathbb{Z}^T\mathbb{Z})(\underline{\hat{\beta}} - \underline{\beta}) \sim \chi^2(r+1)$$

- Bağımsız olarak: $\underline{\hat{\epsilon}}^T\underline{\hat{\epsilon}} \sim \sigma^2\chi^2(n-(r+1))$.
- Bu ikisinin oranı (her biri kendi serbestlik derecesine bölünür) bir F-dağılımı verir ve σ^2 *iptal olur*:

$$(\underline{\hat{\beta}} - \underline{\beta})^T(\mathbb{Z}^T\mathbb{Z})(\underline{\hat{\beta}} - \underline{\beta}) \frac{1}{S^2(r+1)} \sim F(r+1,\ n-(r+1))$$

- **Pivotal miktar:** Bilinen bir dağılıma sahip, bilinmeyen parametre içermiyor. Tüm çıkarım buradan türetilir.

8. Güven Bölgeleri ve Aralıkları

$\underline{\beta}$ için Güven Bölgesi

- $\underline{\hat{\beta}}$ etrafında bir elips, $1-\alpha$ güven bölgesi:

$$CR_{1-\alpha}(\underline{\beta}) = \left\{\underline{\beta} \in \mathbb{R}^{r+1} : \frac{1}{S^2}(\underline{\hat{\beta}} - \underline{\beta})^T(\mathbb{Z}^T\mathbb{Z})(\underline{\hat{\beta}} - \underline{\beta}) \leq (r+1)F_\alpha(r+1,\ n-(r+1))\right\}$$

- Zamanın $(1-\alpha) \times 100\%$'inde gerçek $\underline{\beta}$ bu elipsin içindedir.

σ^2 için Güven Aralığı

- $(n-(r+1))S^2/\sigma^2 \sim \chi^2(n-(r+1))$ olduğundan:

$$CI_{1-\alpha}(\sigma^2) = \left[(n-(r+1))S^2\frac{1}{\chi^2_{\alpha/2}(n-(r+1))},\quad (n-(r+1))S^2\frac{1}{\chi^2_{1-\alpha/2}(n-(r+1))}\right]$$

$\underline{a}^T\underline{\beta}$ için Eş Zamanlı Güven Aralığı

- Herhangi bir doğrusal kombinasyon $\underline{a}^T\underline{\beta}$, $\underline{a} \in \mathbb{R}^{r+1}$ için **Maksimum Lemma** kullanılır:

$$\max_{\underline{a}} \frac{1}{S^2}\frac{(\underline{a}^T(\underline{\hat{\beta}} - \underline{\beta}))^2}{\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}} = \frac{1}{S^2}(\underline{\hat{\beta}} - \underline{\beta})^T(\mathbb{Z}^T\mathbb{Z})(\underline{\hat{\beta}} - \underline{\beta}) \sim (r+1)F(r+1, n-(r+1))$$

- Eş zamanlı güven aralığı:

$$SimCI_{1-\alpha}(\underline{a}^T\underline{\beta}) = \left\{\underline{a}^T\underline{\hat{\beta}} \pm \sqrt{\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}} \cdot \sqrt{S^2(r+1)F_{1-\alpha}(r+1,\ n-(r+1))}\right\}$$